

NPTEL Course on "Computer System Optimization for ML"

Lecture #50

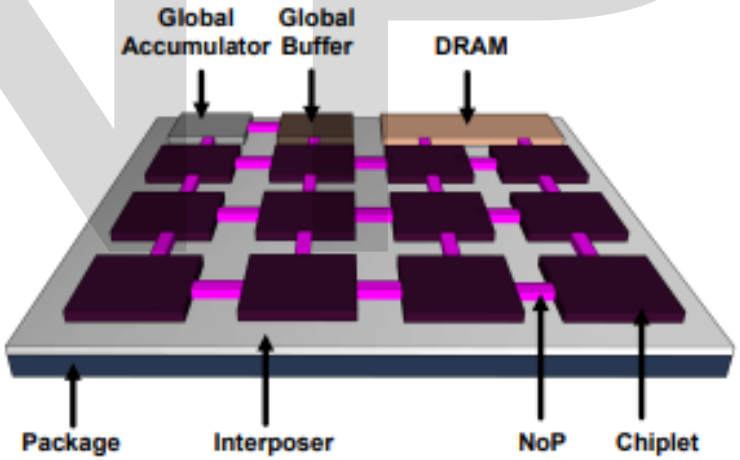
Agenda

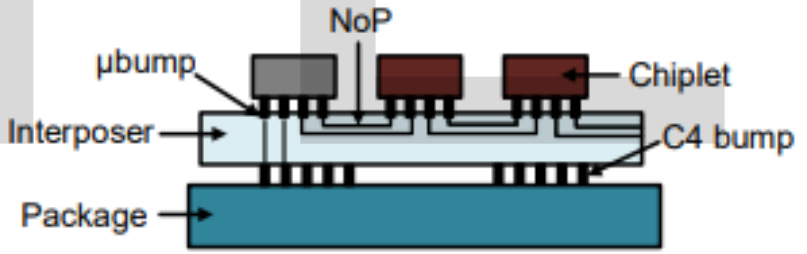
- Chiplet-based Accelerator (contd.)
- Chiplet-based Heterogeneous Accelerator for LLMs

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Background (2.5D Technology)

- Yield for monolithic IC is low
- 3D IC shows thermal hotspots due to increasing number of through-silicon vias (TSV)
- 2.5D technology has emerged recently which increases yield without much of





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Performance Estimation of Computation Elements

- **Modelling-based**
- **Analytical models for low-level circuit components via Predictive Technology Model (PTM) [3]**
  - Mux
  - Decoder
  - ADC/DAC
- **Required number of components are calculated**
  - Performance is estimated via scaling
- **Fast; since equations are executed again and again**

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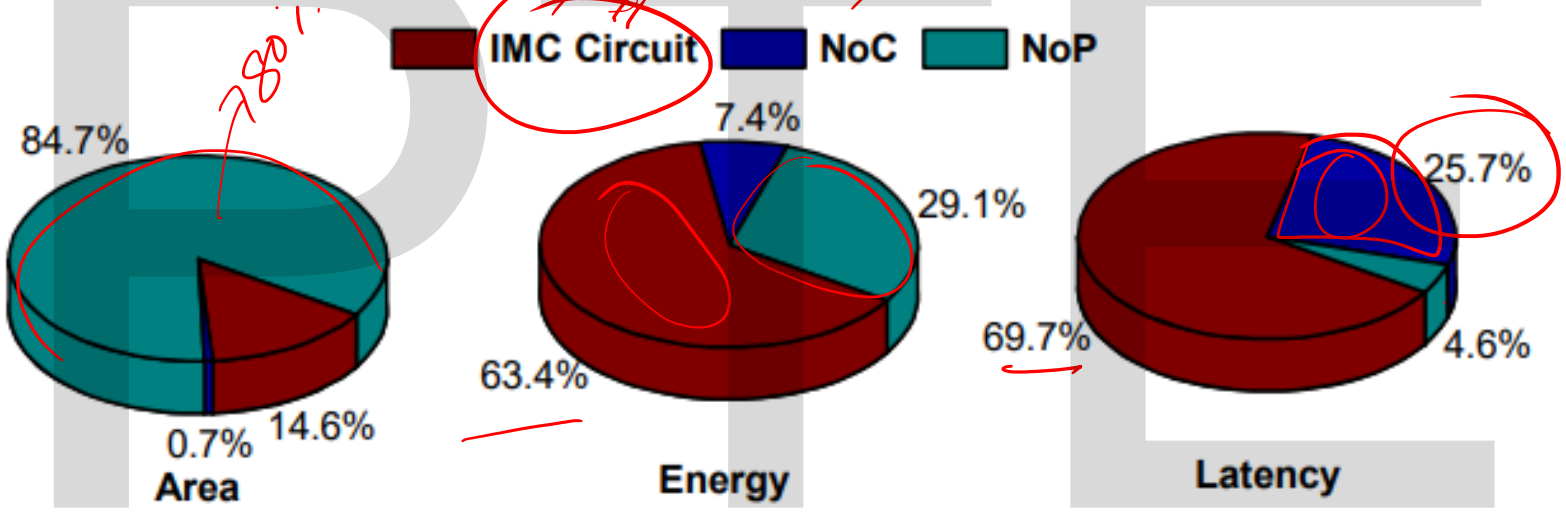
[3] Zhao, Wei, and Yu Cao. "Predictive technology model for nano-CMOS design exploration." ACM Journal on Emerging Technologies in Computing Systems (JETC) 3.1 (2007): 1-es.

Performance Estimation of Communication

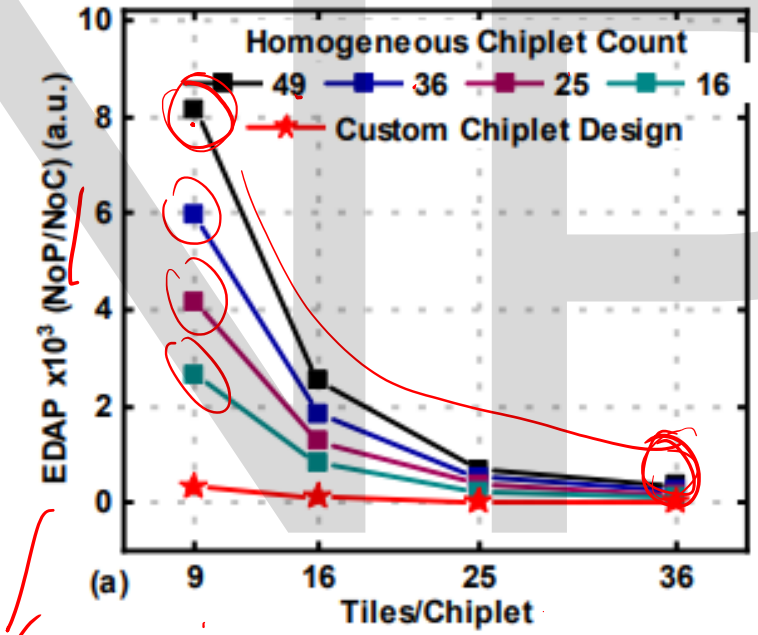
- **Typically, a block of computation is performed at once**
- **Many packets are ready to be communicated at a single cycle**
  - No particular distribution in arrival rate
- **Need to know the total latency**
  - Time taken by the communication to be finished
  - Average latency is not important
- **Deploying queuing theory-based analytical model is not feasible**
- **Cycle accurate simulation is performed to estimate communication performance**

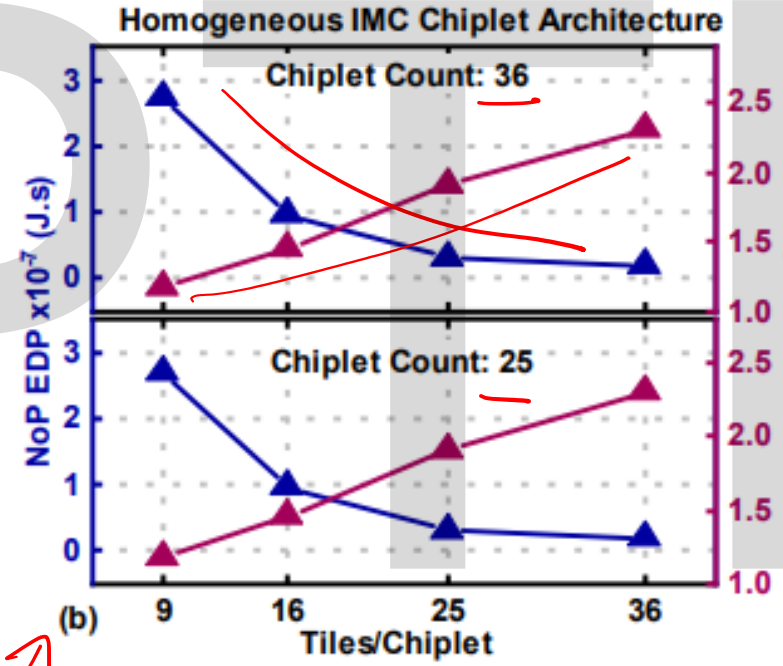
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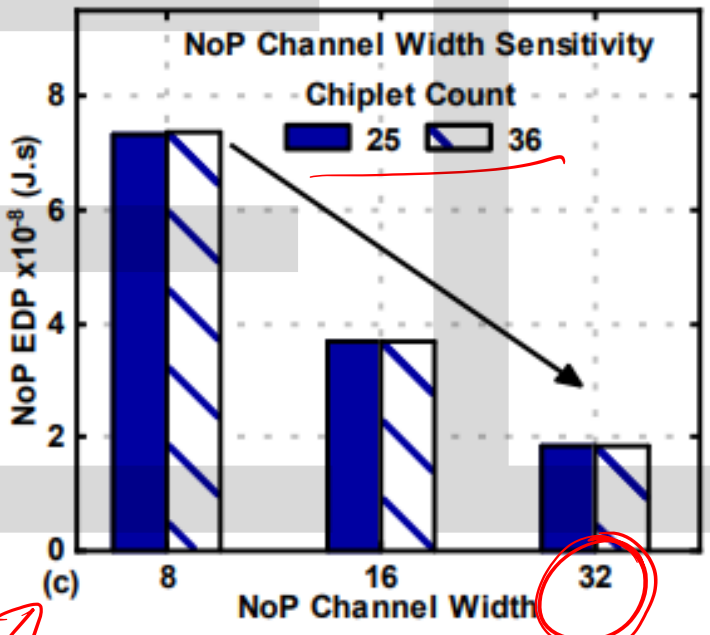
Results (1)


IMC Circuit
NoC
NoP
84.7%
0.7%
14.6%
Area
7.4%
29.1%
63.4%
Energy
25.7%
69.7%
4.6%
Latency

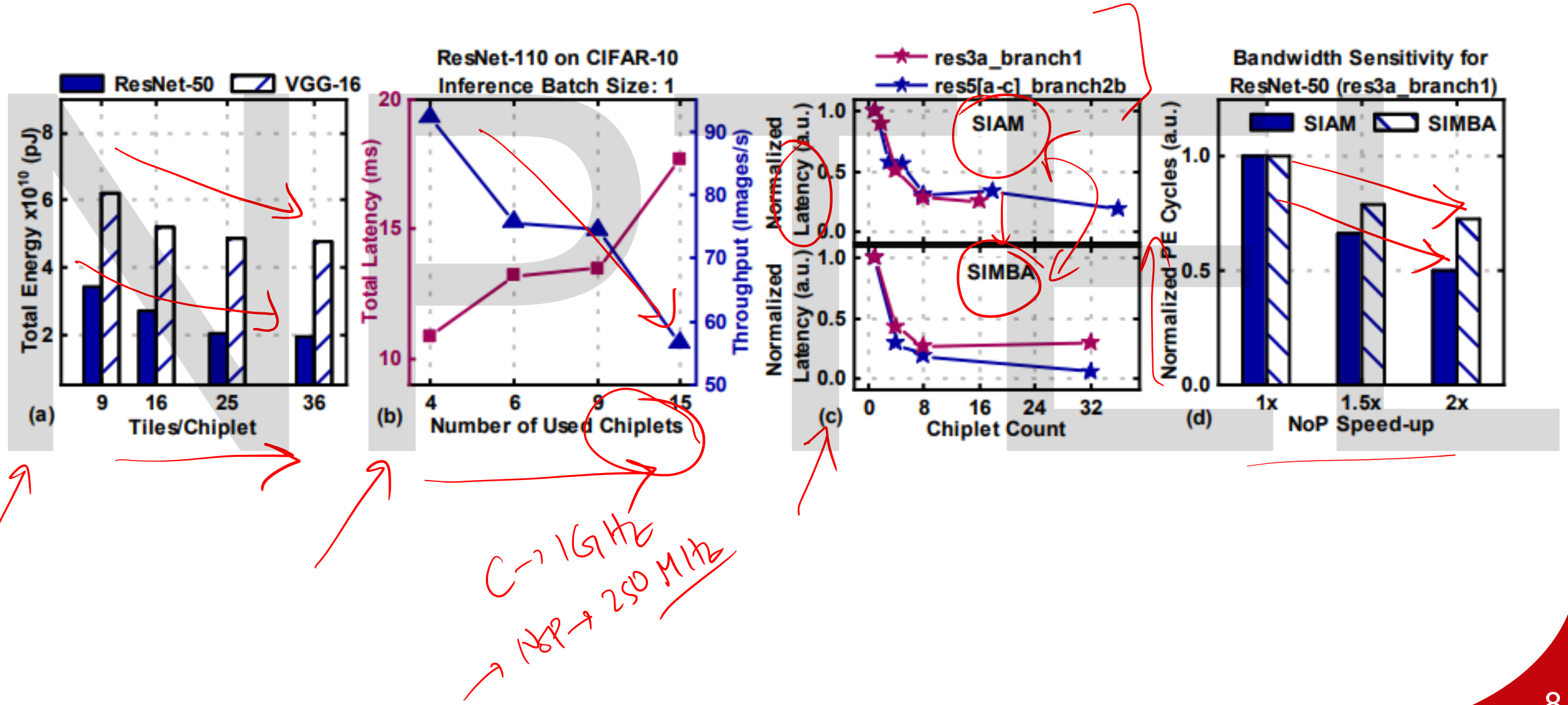
Results (2)


Homogeneous Chiplet Count
49
36
25
16
Custom Chiplet Design
EDAP x10³ (NoP/NoC) (a.u.)
Tiles/Chiplet
(a)


Homogeneous IMC Chiplet Architecture
Chiplet Count: 36
Chiplet Count: 25
NoP EDP x10⁻⁷ (J.s)
NoC EDP x10⁻⁸ (J.s)
Tiles/Chiplet
(b)


NoP Channel Width Sensitivity
Chiplet Count
25
36
NoP EDP x10⁻⁸ (J.s)
NoP Channel Width
(c)

Results (3)


ResNet-50
VGG-16
Total Energy x10^10 (pJ)
Tiles/Chiplet
(a)
ResNet-110 on CIFAR-10
Inference Batch Size: 1
Total Latency (ms)
Throughput (Images/s)
Number of Used Chiplets
(b)
res3a_branch1
res5[a-c]_branch2b
Normalized Latency (a.u.)
SIAM
SIMBA
Chiplet Count
(c)
Bandwidth Sensitivity for
ResNet-50 (res3a_branch1)
SIAM
SIMBA
Normalized PE Cycles (a.u.)
NoP Speed-up
(d)

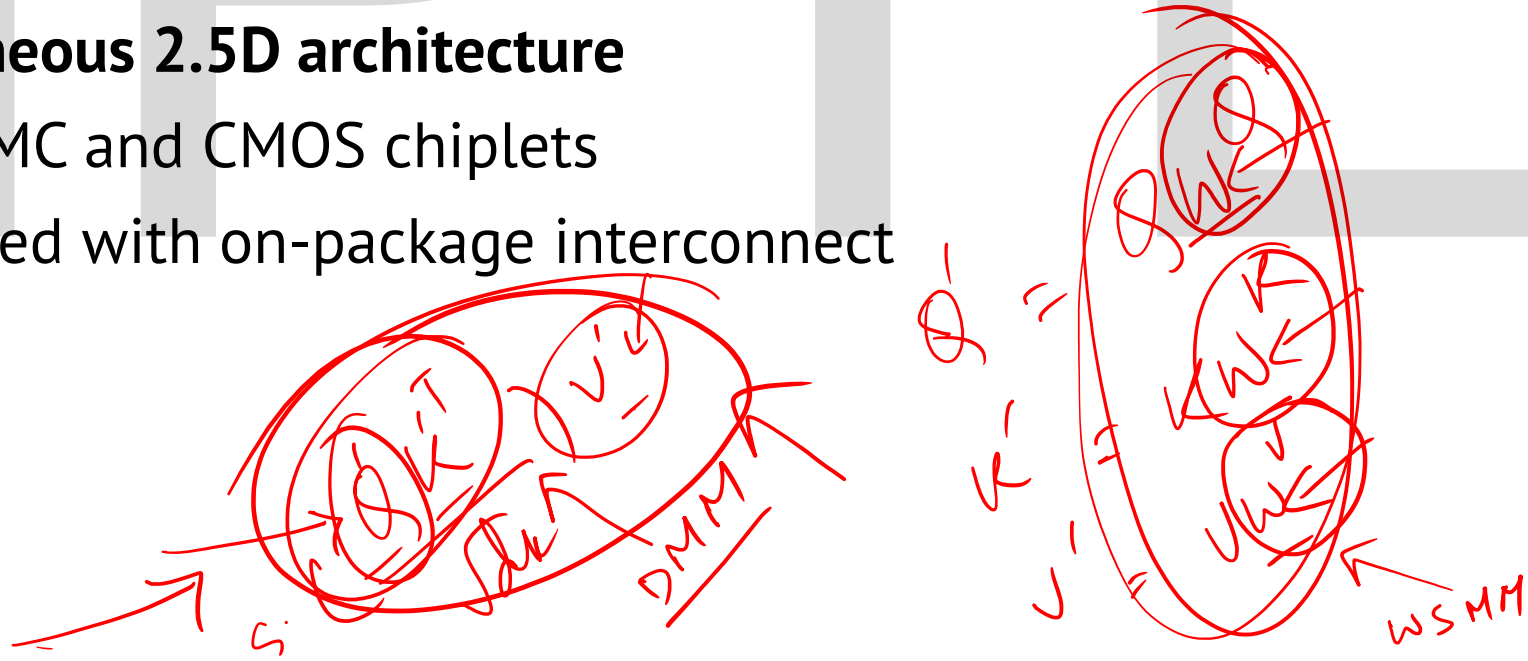
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- Chiplet-based Accelerator (contd.)
- **Chiplet-based Heterogeneous Accelerator for LLMs**

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Need for Heterogeneity

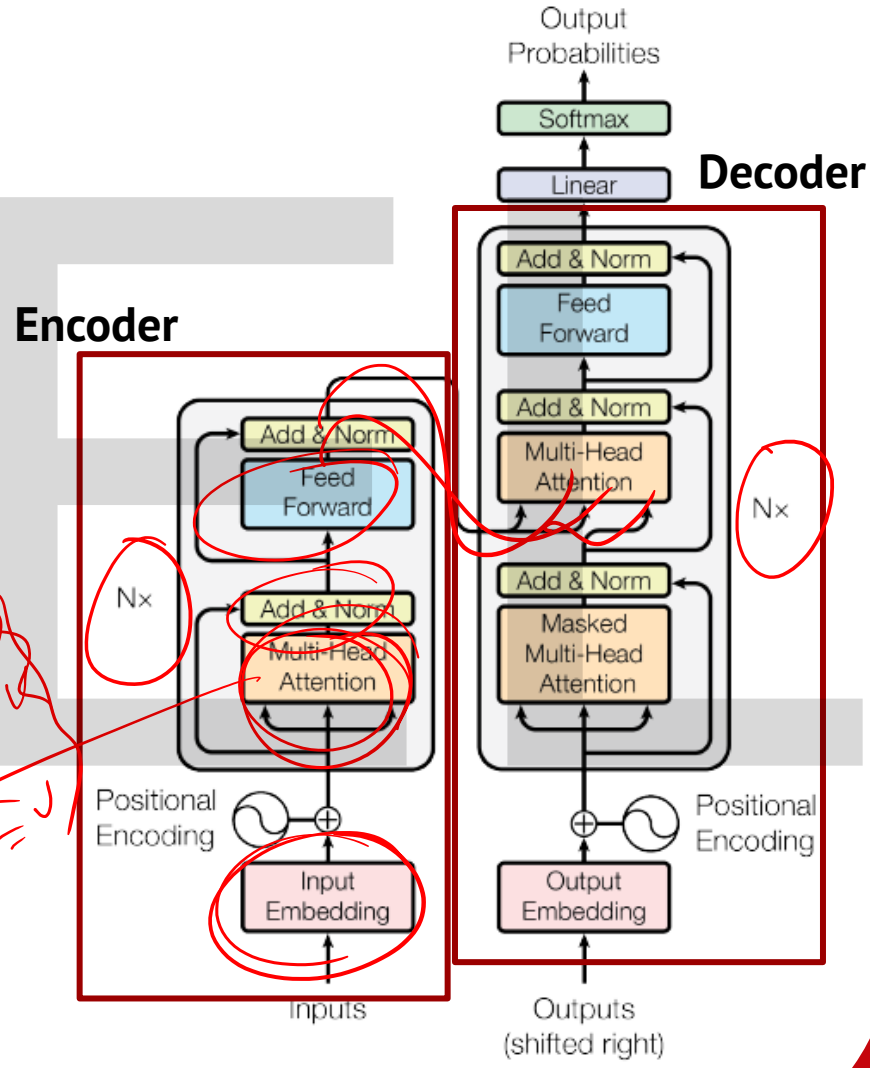
- **IMC chiplets**
  - Suitable for weight-stationary multiplications (Q, K, V)
  - Not suitable for multiplications where both inputs are dynamic
  - High write latency/energy
- **CMOS chiplets are suitable for dynamic matrix multiplication**
- **Heterogeneous 2.5D architecture**
  - Mix of IMC and CMOS chiplets
  - Connected with on-package interconnect



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LLM Basics

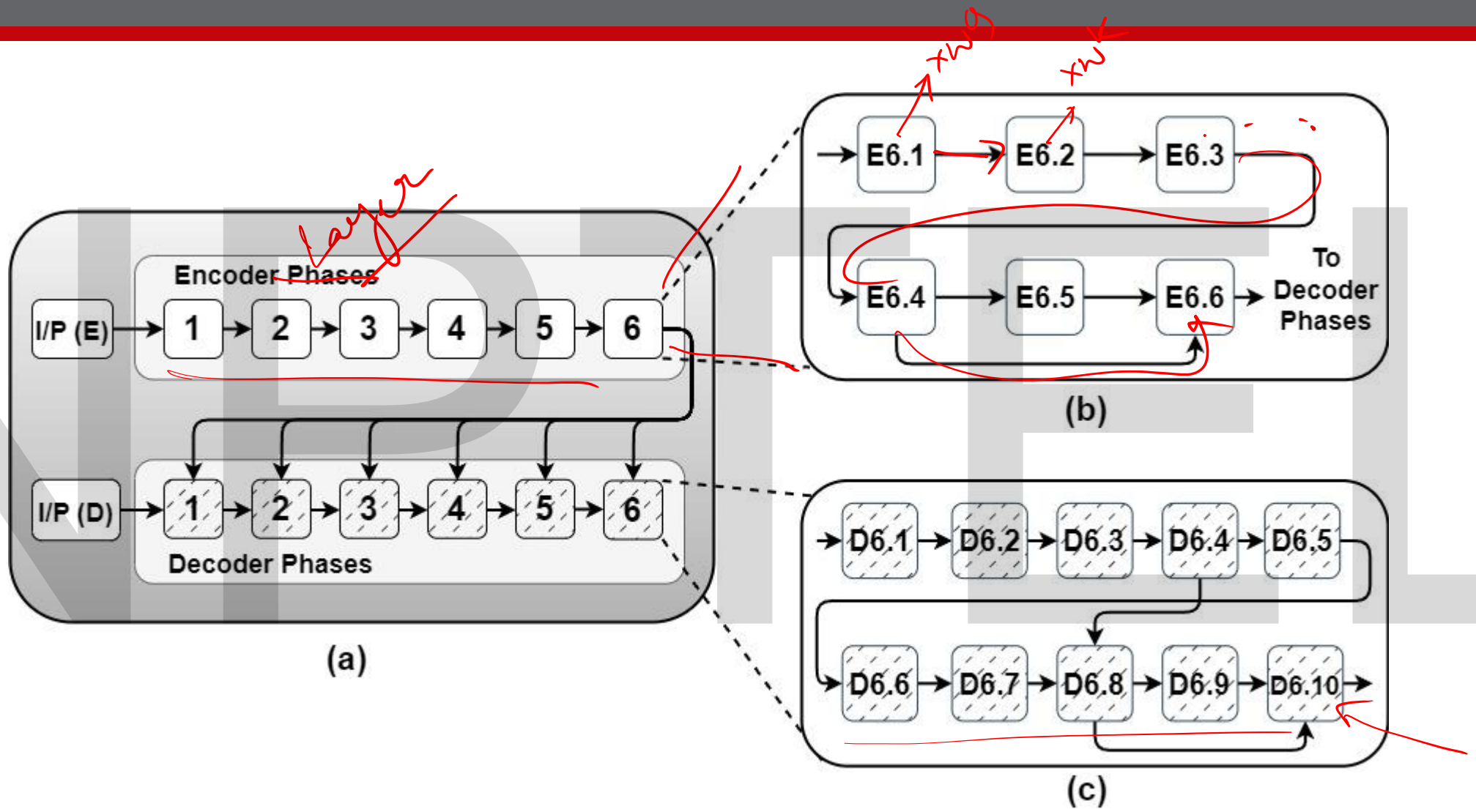
- **Based on Encoders and Decoders**
  - **They are connected**
- **Input to Encoder is a sequence of words (sentence)**
- **Input to Decoder**
  - **Outputs (shifted right)**
  - **Output probabilities**
  - **Input from Encoder**



[4] Vaswani, Ashish, et al. "Attention is all you need." *Advances in neural information processing systems* 30 (2017).

NOTE

Simpler View of LLMs





THANK YOU

Slide Courtesy: Murali Annavaram (University of Southern California), R. Govindarajan (IISc), Arkaprava Basu (IISc), Jayashree Mohan (Microsoft Research)